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Kinetics of Biosynthesis of Acetylcholine Receptor and Subsequent Incorporation into Plasma Membrane of Cultured Chick Skeletal Muscle

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Summary

20% of the acetylcholine receptors in cultured chick skeletal muscle remain unbound following long-term growth of muscle in medium containing a potent, essentially irreversible receptor-blocking agent, α-bungarotoxin. About half the receptors which are unavailable for interaction with extracellular α-bungarotoxin are newly synthesized molecules which presumably are being processed and transported to the plasma membrane. When the muscle cultures are switched to a medium containing ^{2}H, ^{13}C, ^{15}N-amino acids, these receptors are rapidly labeled, the fraction of labeled molecules beginning to plateau at 3 hr. Few labeled receptors appear in the plasma membrane during the first 3 hr of labeling with ^{2}H, ^{13}C, ^{15}N-amino acids. After 3.5 hr of labeling, virtually all the receptors being incorporated into the plasma membrane are labeled receptors. The kinetics of labeling of the "pool" and "surface" receptors with ^{2}H, ^{13}C, ^{15}N-amino acids confirm the "precursor-product" type relationship of pool and surface acetylcholine receptors.

In this study, receptors synthesized in medium containing ^{2}H, ^{13}C, ^{15}N-amino acids were resolved from ^{1}H, ^{12}C, ^{14}N-receptors by velocity sedimentation in sucrose-deuterium oxide and sucrose-H_2O gradients, and their densities were estimated from sedimentation rates in shallow gradients of various average density. Estimated densities were 1.32 g/cm^3 for ^{1}H, ^{12}C, ^{14}N-receptors and 1.41 g/cm^3 for ^{2}H, ^{13}C, ^{15}N-receptors. This density difference corresponds to 80% substitution of normal aminoacyl residues by ^{2}H, ^{13}C, ^{15}N-residues in the denser receptor.

Introduction

In previous reports, the kinetics of turnover of acetylcholine receptors in membranes of developing chick muscle fibers have been described and the results of some inhibitor studies have been presented (Hartzell and Fambrough, 1973; Devreotes and Fambrough, 1975, 1976a, 1976b). This report focuses on one aspect of receptor metabolism—intracellular processing of newly synthesized receptors.

A model has been presented which describes the biosynthesis and degradation of acetylcholine receptors (Devreotes and Fambrough, 1976b). This model is based upon studies which suggest that receptors are synthesized de novo in the myotubes and undergo a 3 hr transit before exposure on the myotube surface. This intracellular transport process is independent of protein synthesis, but is energy-dependent and temperature-dependent. Puromycin and cycloheximide have little effect on the appearance of new receptors at the surface for 2-3 hr, but after this initial lag, become effective inhibitors. 2, 4-dinitrophenol (DNP) and carbonyl-M-chlorophenyl hydrazone (FCCP) block incorporation of new receptors into the surface membrane at 37°C from the outset. Similarly, incorporation of new receptors cannot proceed at low temperatures.

The observation that new receptors continue to appear at the surface for 2-3 hr in the absence of protein synthesis suggested that a pool of presynthesized receptors exists. The "pool receptors" were identified in nonionic detergent extracts of myogenic cell cultures (Devreotes and Fambrough, 1975). All the soluble α-bungarotoxin binding sites considered to represent pool receptors were d-tubocurarine-competed and had a sedimentation coefficient of about 10S. The number of these α-bungarotoxin binding sites, or receptors, decreased rapidly when muscle cultures were treated with puromycin. The kinetics of decrease of pool receptors approximately matched the kinetics of appearance of new surface receptors in the presence of puromycin—that is, the pool was significantly depleted at the time new receptors ceased to appear at the surface.

We previously reported that acetylcholine receptors synthesized by embryonic chick myotubes grown in medium containing ^{2}H or ^{13}C-amino acids can be separated from the usual ^{1}H, ^{12}C-receptors on the basis of the difference in bouyant density (Devreotes and Fambrough, 1976a). The density-shift technique has been improved by use of ^{2}H, ^{13}C, ^{15}N-amino acids. ^{2}H, ^{13}C, ^{15}N-receptors, detected as ^{125}I-mono-iodo-α-bungarotoxin-receptor complexes, can be resolved from normal receptors, detected as ^{131}I-mono-iodo-α-bungarotoxin-receptor complexes, by velocity gradient sedimentation. This technique makes possible the detection of newly synthesized acetylcholine receptors

in crude nonionic detergent extracts of embryonic myotubes. In this report, improvements in the density-shift technique are described, and an estimate of the magnitude of the density shift for acetylcholine receptors is offered. This technique is used to demonstrate directly the existence of a pool of recently synthesized acetylcholine receptors in chick myotubes, and to describe the kinetics of receptor biosynthesis and incorporation into the plasma membrane.

Results

Separation of ^{2}H, ^{13}C, ^{15}N- and ^{1}H, ^{12}C, ^{14}N-Receptors and Estimation of the Density Shift

A strategy was devised so that chick myogenic cells would differentiate in the presence of ^{2}H, ^{13}C, ^{15}N-amino acids. Myogenic cells were seeded on 100 mm diameter culture dishes in normal medium and allowed to proliferate for 1 day. The cultures were then rinsed twice with sterile Hank's balanced salt solution to remove completely the ^{1}H, ^{12}C, ^{14}N-amino acid-containing medium and fed with ^{2}H, ^{13}C, ^{15}N-amino acid-containing medium. Cells were grown for 2 additional days, and at the end of this incubation period, surface acetylcholine receptors were labeled with 0.2 μg/ml ^{125}I-mono-iodo-α-bungarotoxin for 30 min at 37°C. The cultures were rinsed to remove unbound α-bungarotoxin and extracted into nonionic detergents as described in Experimental Procedures. The myogenic cells were able to differentiate in the presence of the ^{2}H, ^{13}C, ^{15}N-amino acids, and large numbers of α-bungarotoxin binding sites were present at the end of the incubation period.

^{125}I-mono-iodo-α-bungarotoxin-^{2}H, ^{13}C, ^{15}N-acetylcholine receptor complexes were mixed with a "marker" of ^{131}I-mono-iodo-α-bungarotoxin-^{1}H, ^{12}C, ^{14}N-receptor complexes prepared from myotubes which had differentiated in ^{1}H, ^{12}C, ^{14}N-amino acid-containing medium. The mixed extract was then analyzed by velocity sedimentation in 25–40% sucrose-deuterium oxide gradients. As illustrated in Figure 1, the ^{125}I-mono-iodo-α-bungarotoxin-^{2}H, ^{13}C, ^{15}N-receptor complexes are well separated from the ^{131}I-mono-iodo-α-bungarotoxin-^{1}H, ^{12}C, ^{14}N-receptor complexes. When the ^{125}I- and ^{131}I-mono-iodo-α-bungarotoxin labels are switched, the results are unaffected—that is, ^{131}I-mono-iodo-α-bungarotoxin-^{2}H, ^{13}C, ^{15}N-receptor complexes are equally well separated from ^{125}I-mono-iodo-α-bungarotoxin-^{1}H, ^{12}C, ^{14}N-receptor complexes. These results indicate that a large fraction of the newly appearing acetylcholine receptors which occur on myogenic cells as they differentiate are the result of de novo receptor synthesis.

It was important to determine to what extent the newly synthesized receptors had been substituted with ^{2}H, ^{13}C, ^{15}N-amino acids. The size of the density shift was estimated as described in Experimental Procedures and compared with the theoretically expected shift. Mixed extracts of ^{125}I-mono-iodo-α-bungarotoxin-^{2}H, ^{13}C, ^{15}N-receptor complexes and ^{131}I-mono-iodo-α-bungarotoxin-^{1}H, ^{12}C, ^{14}N-receptor complexes, prepared as described for Figure 1, were centrifuged for various lengths of time in the H_2O, 50% D_2O, and D_2O sucrose gradients. The distance traveled by each type of α-bungarotoxin-receptor complex was determined, and the results are shown in Figure 2. As explained in Ex-

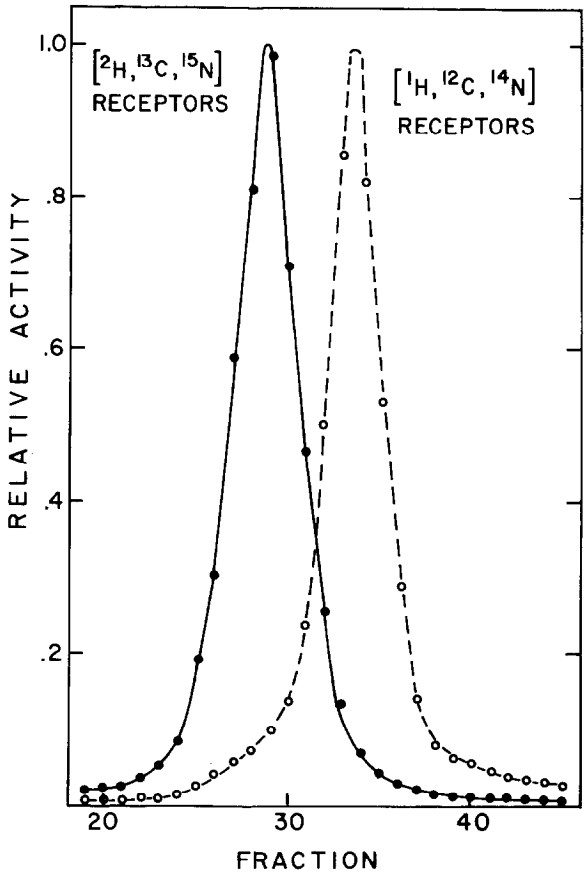

Figure 1. Analysis of Acetylcholine Receptors Produced by Myotubes Differentiating in Vitro in the Presence of ^{2}H, ^{13}C, ^{15}N-Amino Acids

Myogenic cells were grown in tissue culture from 18–66 hr in medium containing the isotopically substituted amino acids. After 66 hr in culture, surface acetylcholine receptors on the myotubes were saturated with ^{125}I-mono-iodo-α-bungarotoxin, the unbound toxin was removed, and the receptors were solubilized in detergent solution. ^{131}I-mono-iodo-α-bungarotoxin-^{1}H, ^{12}C, ^{14}N-receptors were added to the sample. Velocity sedimentation was performed as described in Experimental Procedures. Approximately 0.2 ml fractions were collected, and only a portion of the 12 ml gradient is shown.

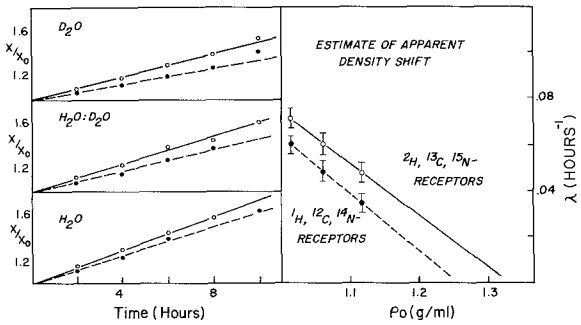

Figure 2. Simultaneous Estimates of the Densities of ^1H, ^{12}C, ^{14}N- and ^2H, ^{13}C, ^{15}N-Receptors

Velocity sedimentation of mixtures of the mono-iodo-α-bungarotoxin-receptor complexes in shallow gradients of various average densities was performed as described in Experimental Procedures.

Left panel: distance traveled by each type of mono-iodo-α-bungarotoxin-receptor complex as a function of time during velocity sedimentation.

Right panel: slopes of the curves taken from the left panel plotted against ρ_0, the average density of each type of gradient.

(●- - -●) ^1H, ^{12}C, ^{14}N-receptors; (○——○) ^2H, ^{13}C, ^{15}N-receptors.

perimental Procedures, the extrapolated density at which the velocity of sedimentation equals zero is an estimate of the density of the macromolecule. The apparent density of mono-iodo-α-bungarotoxin-^1H, ^{12}C, ^{14}N-receptor complexes is about 1.26 g/cm^3, while the apparent density of mono-iodo-α-bungarotoxin-^2H, ^{13}C, ^{15}N-receptor complexes is about 1.33 g/cm^3. Bound detergent accounts in part for the anomolous low density of the complexes. Meunier, Olsen, and Changeux (1972a) estimated that bound detergent accounts for about 20% of the weight of purified, solubilized receptors from Electrophorus electricus. We assumed that bound detergent would make a similar contribution to chick receptors and would thereby partially obscure the density shift. When corrections were made for detergent binding, the apparent densities were 1.32 and 1.40 g/cm^3, respectively. Taking into account the contribution of bound α-bungarotoxin (ρ = 1.33 g/cm^3) to the densities of the complexes (assuming 1 mole of α-bungarotoxin molecules bound per 100,000 g of receptor protein; see Karlin, 1974), these values become 1.32 and 1.41 g/cm^3.

The estimated apparent density shift between ^2H, ^{13}C, ^{15}N- and ^1H, ^{12}C, ^{14}N-receptors is about 0.09 g/cm,^{3} which corresponds to a shift of about 6.8%. The maximum theoretically expected shift is about 0.11 g/cm^3 or about 8.5% (Hu, Bock, and Halvorson, 1962). The same maximal shift was calculated from the amino acid composition of the eel acetylcholine receptor determined by Meunier et al. (1972b). Thus a substantial fraction of the amino acids of the receptor—about 80%—have been isotopically substituted by the ^2H, ^{13}C, ^{15}N- amino acids. In the kinetic studies described below, we found that the data on the time course of shift in receptor density were compatible with a rapid equilibration of ^2H, ^{13}C, ^{15}N-amino acids with the intracellular amino acid pools, resulting in the biosynthesis of maximally dense receptors within a few minutes.

Kinetics of Receptor Biosynthesis and Incorporation into Plasma Membrane

The density-shift technique was used to measure directly the intracellular transit time between synthesis of new receptors and their appearance at the myotube surface. Existing surface receptors were blocked by overnight exposure of myotubes to unlabeled α-bungarotoxin. Unbound α-bungarotoxin was rinsed away, and cultures were incubated in medium containing ^2H, ^{13}C, ^{15}N- amino acids. All the newly appearing surface receptors were labeled by a 30 min exposure to ^{125}I-mono-iodo-α-bungarotoxin. These complexes were extracted into nonionic detergents and were mixed with a "marker" of ^{131}I-mono-iodo-α-bungarotoxin-^1H, ^{12}C, ^{14}N-receptor complexes. The mixed extract was analyzed on 25–40% sucrose-deuterium oxide gradients. The fraction of ^{125}I-mono-iodo-α-bungarotoxin-receptor complexes detected in the position of α-bungarotoxin-^2H, ^{13}C, ^{15}N-receptor complexes was considered to be the fraction of newly appearing surface receptors which were synthesized after the start of the experiment. Similarly, the fraction detected in the position of α-bungarotoxin-^1H, ^{12}C, ^{14}N-receptors was considered to consist of presynthesized receptors. The kinetics of incorporation of total and ^2H, ^{13}C, ^{15}N-receptors are shown in Figure 3. Several examples of gradients used to construct the kinetics curve are shown in Figure 4. It is clear from Figures 3 and 4 that most of the receptors appearing on the surface during the first 3 hr after introduction of the isotopic label are ^1H, ^{12}C, ^{14}N-receptors. Small amounts of ^2H, ^{13}C, ^{15}N-receptors appear at 2 and 3 hr; after slightly more than 3 hr, most of the newly incorporated receptors are detected as ^2H, ^{13}C, ^{15}N-receptors. Thus upon introduction of the isotopically labeled amino acids, a lag of slightly more than 3 hr occurs before isotopically labeled receptors rapidly appear at the surface.

In contrast to the lag before appearance of ^2H, ^{13}C, ^{15}N-receptors at the myotube surface, ^2H, ^{13}C, ^{15}N-receptors rapidly enter the intracellular receptor pool. Large cultures of chick myotubes were equilibrated in medium containing α-bungarotoxin. At subsequent times, sets of cultures were switched to medium containing ^2H, ^{13}C, ^{15}N-amino acids and still containing unlabeled α-bungarotoxin. Thus each set was exposed to the ^2H, ^{13}C, ^{15}N-amino acids for different times, but all the cul-

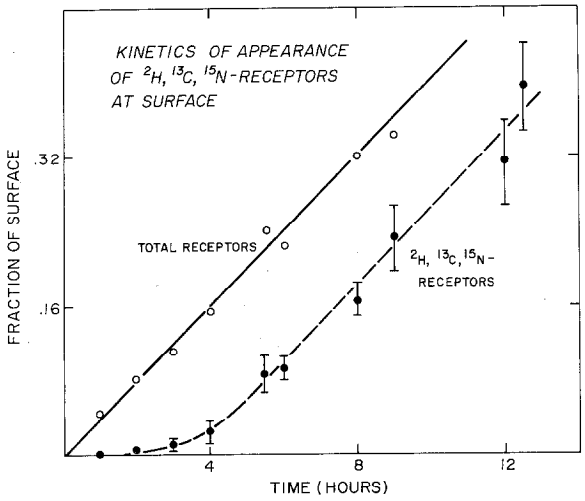

Figure 3. Kinetics of Appearance of Total (^{1}H, ^{12}C, ^{14}N- plus ^{2}H, ^{13}C, ^{15}N-) and Density-Shifted (^{2}H, ^{13}C, ^{15}N-) Receptors on Myotube Surfaces during Incubation in Medium Containing ^{2}H, ^{13}C, ^{15}N-Amino Acids

After long-term incubation in medium containing unlabeled α-bungarotoxin, cultures were rinsed to remove unbound α-bungarotoxin and cultured in medium containing ^{2}H, ^{13}C, ^{15}N-amino acids. Receptors appearing on the myotube surfaces were labeled by addition of 0.15 μg/ml ^{125}I-mono-iodo-α-bungarotoxin to the culture medium during the final 30 min. Unbound ^{125}I-mono-iodo-α-bungarotoxin was removed by rinsing cultures in the cold, and the ^{125}I-mono-iodo-α-bungarotoxin-receptor complexes were extracted and analyzed by sucrose gradient velocity sedimentation as described in Experimental Procedures. The receptors from 3–5 100 mm diameter cultures were used for each data point. Data from 2 experiments were combined by first subtracting the small background present at zero time and then normalizing the slope for total receptor appearance to 4% addition of new receptors to the total surface plus hidden receptor population per hour, a typical rate of incorporation. The fraction of ^{2}H, ^{13}C, ^{15}N- receptors at each time point was calculated from areas under curves on sucrose gradients (see Figure 4 and Experimental Procedures). (○——○) total receptors; (●- - -●) ^{2}H, ^{13}C, ^{15}N-receptors.

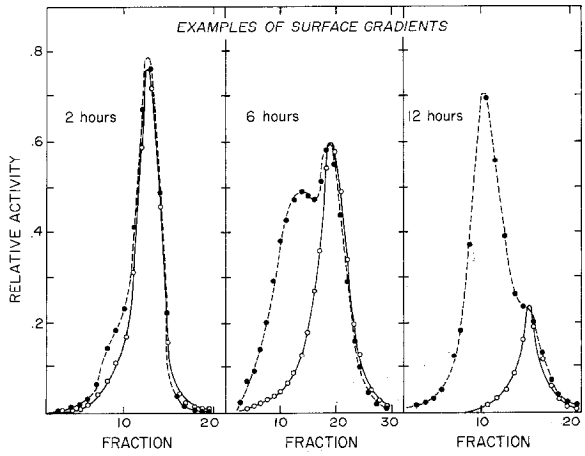

Figure 4. Sucrose Gradient Velocity Sedimentation Profiles of Surface Receptors Appearing during Culture of Myogenic Cells in Medium Containing ^{2}H, ^{13}C, ^{15}N-Amino Acids

Gradient profiles illustrated here are those from which the data for the 2, 6, and 12 hr time points in Figure 3 were obtained. A marker of ^{131}I-mono-iodo-α-bungarotoxin-^{1}H, ^{12}C, ^{14}N- receptor complexes was mixed with the sample before centrifugation, and its position in the gradient is identical to the position of the ^{125}I-mono-iodo-α-bungarotoxin-^{1}H, ^{12}C, ^{14}N-receptor complexes. Note the small number of ^{2}H, ^{13}C, ^{15}N-receptors present at 2 hr.

tures were incubated with unlabeled α-bungarotoxin for about 10 hr. At the end of the incubation period, pool receptors were prepared and labeled with ^{125}I-mono-iodo-α-bungarotoxin as described in Experimental Procedures. Within experimental error (standard deviation= $\pm$9.36% of mean, n = 30), an identical number of "pool" receptors was present in the extract from each culture set. The ^{125}I-mono-iodo-α-bungarotoxin-receptor complexes were mixed with a "marker" of ^{131}I-mono-iodo-α-bungarotoxin-^{1}H, ^{12}C, ^{14}N-receptor complexes and analyzed on 25–40% sucrose-deuterium oxide gradients. The fraction of pool complexes located in the position of ^{2}H, ^{13}C, ^{15}N-receptor was determined. The kinetics of "switchover" of the pool from ^{1}H, ^{12}C, ^{14}N-receptors to ^{2}H, ^{13}C, ^{15}N-receptors are shown in Figure 5. Several representative gradients are shown in Figure 6. After introduction of the isotopically labeled amino acids, ^{1}H, ^{12}C, ^{14}N-receptors disappear rapidly from the pool (by entry into the surface). These receptors are rapidly replaced by ^{2}H, ^{13}C, ^{15}N-receptors, and the total number of pool receptors remains constant. During the initial rapid phase of the "switchover", about 50% of the presynthesized pool receptors are replaced by newly synthesized ones. The remaining 50% of the ^{1}H, ^{12}C, ^{14}N-presynthesized pool receptors are only slowly replaced by ^{2}H, ^{13}C, ^{15}N-receptors.

Pool Size Determination

As described in Experimental Procedures, our in vitro reaction to label pool receptors has been considerably modified to reduce the amount of nonspecific binding. By use of these methods, we measured the pool size more precisely than had been done previously. The total number of ^{125}I-mono-iodo-α-bungarotoxin binding sites found in cultures incubated with unlabeled α-bungarotoxin overnight was compared to the number found in unpretreated cultures. One set of cultures was incubated with unlabeled α-bungarotoxin for 12 hr, while an identical set received no pretreatment with α-bungarotoxin. Thereafter, both sets were handled identically—cultures were rinsed, soluble extracts were prepared, and in vitro reactions were carried out as described in Experimental Procedures. After removal of unbound ^{125}I-mono-iodo-α-bungarotoxin by BioGel P60 chromatography, soluble receptors were analyzed by sucrose gradient velocity sedimentation. 10S peaks from pretreated cultures (pool) contained about 15–18% as much radioactivity as 10S peaks from unpretreated

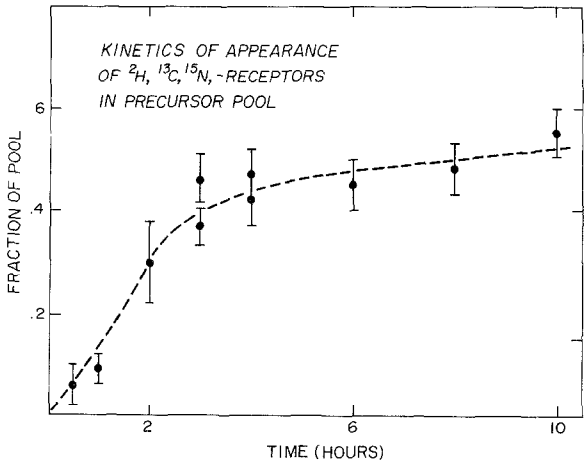

Figure 5. Kinetics of Appearance of ^{2}H, ^{13}C, ^{15}N-Receptors in the Intracellular Pool

^{125}I-mono-iodo-α-bungarotoxin-receptor complexes involving the pool receptors were prepared as described in Experimental Procedures from cultures incubated for the designated times in medium containing ^{2}H, ^{13}C, ^{15}N-amino acids. These receptors were analyzed by sucrose gradient velocity sedimentation to determine the fraction of pool receptors which were ^{2}H, ^{13}C, ^{15}N-receptors.

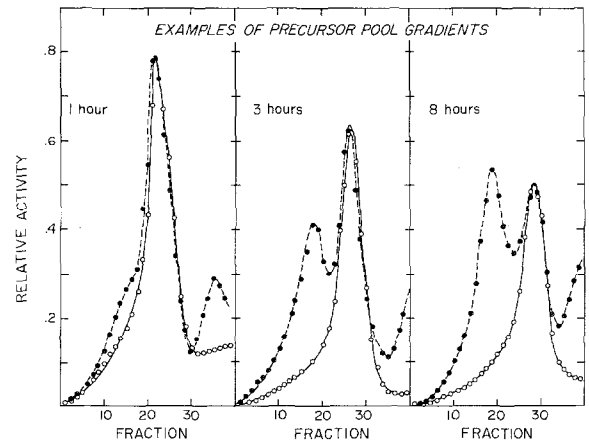

Figure 6. Sucrose Gradient Velocity Sedimentation Profiles of Pool Receptors from Cultures Maintained for Various Periods of Time in Medium Containing ^{2}H, ^{13}C, ^{15}N-Amino Acids

Gradient profiles illustrated here are those from which the data for the 1, 3, and 8 hr time points in Figure 5 were obtained. A marker of ^{131}I-mono-iodo-α-bungarotoxin-^{1}H, ^{12}C, ^{14}N-receptor complexes (prepared as were the complexes containing pool receptors; see Experimental Procedures) was mixed with the sample before centrifugation, and its position in the gradient is identical to the position of the ^{125}I-mono-iodo-α-bungarotoxin-^{1}H, ^{12}C, ^{14}N-receptor complexes.

cultures (pool and total surface). The pool size was calculated to contain about 18–22% as many binding sites as the total surface [surface plus "hidden" classes of receptors in the nomenclature of Devreotes and Fambrough (1975, 1976b)].

Discussion

The Density-Shift Technique

The major advantage of the density-shift technique is as a means of distinguishing newly synthesized molecules of a protein from preexisting ones without the necessity of purification. It is only necessary that the protein of interest can be identified in crude extracts. In a previous publication, metrizamide deuterium-oxide gradients were used to separate α-bungarotoxin-^{2}H-receptor complexes or α-bungarotoxin-^{13}C-receptor complexes from α-bungarotoxin-^{1}H-receptor complexes (Devreotes and Fambrough, 1976a). It has since been found that larger separations can be achieved by velocity sedimentation on sucrose-deuterium oxide gradients. Although sucrose gradients theoretically have lower resolving power than metrizamide gradients, empirically they produce larger separations. This is probably because metrizamide interacts with α-bungarotoxin-receptor complexes (see discussion in Hüttermann and Wendlberger, 1976). We have found that pretreatment of α-bungarotoxin-receptor complexes with high concentrations (50% w/v) of metrizamide causes disruption of the receptors. Thus after such a treatment, the S value of the α-bungarotoxin-receptor complexes is reduced to 4S. Pretreatment with lower concentrations, such as those previously used, did not disrupt receptors; it seemed reasonable, however, that interactions might still exist. Such interactions would obscure detection of density shifts. This report describes several improvements on the density-shift technique. By using ^{2}H, ^{13}C, ^{15}N-amino acids as protein precursors and by changing the gradient sedimentation system from metrizamide to sucrose, larger separations of ^{2}H, ^{13}C, ^{15}N-receptors and ^{1}H, ^{12}C, ^{14}H-receptors were achieved.

Estimates of the size of the density shift suggest that $> 80\%$ of the amino acids of acetylcholine receptors can be substituted by the isotopically labeled amino acids. In deriving this estimate, no correction was made for the contribution of carbohydrate groups. Thus our estimate of about 80% amino acid substitution in the receptor may be an underestimate. The large degree of substitution suggests that most of the subunits of the newly synthesized receptors are synthesized from these amino acids. Since 100% substitution is not achieved, it is possible that there is a continuous, endogenous source of unlabeled amino acids (such as from de novo amino acid synthesis or from protein degradation).

An assumption made in these calculations is that the ^{1}H, ^{12}C, ^{14}N-receptors and ^{2}H, ^{13}C, ^{15}N-receptors are of identical shape and size. If a major distortion in receptor shape accompanied substitution of ^{2}H, ^{13}C, ^{15}N-amino acyl residues for normal ones, a loss or some aberration of receptor func-

tion might be expected. However, receptor function was not measurably affected as judged by electrophysiological analyses: using intracellular recording and iontophoretic application of acetylcholine ions onto myotubes, we found rapid responses and high sensitivity to acetylcholine mediated by ^{2}H, ^{13}C, ^{15}N-receptors.

^{2}H, ^{13}C, ^{15}N-amino acids were supplied to embryonic myogenic cells 18–66 hr after plating, during the critical time of differentiation. The myogenic cells are able to differentiate in the presence of the ^{2}H, ^{13}C, ^{15}N-amino acids, and large numbers of acetylcholine receptors appear on the myotube surfaces. As shown in Figure 1, nearly all these receptors are ^{2}H, ^{13}C, ^{15}N-receptors, and therefore synthesized sometime after the myogenic cells had been in culture for 18 hr. This result suggests there are no significant stores of acetylcholine receptors in myoblasts, which later in myogenesis appear on the cell surface.

Kinetics of Receptor Biosynthesis and Incorporation into Plasma Membranes

The observations reported here confirm and extend earlier observations on the intracellular transport of acetylcholine receptors in developing chick myotubes. Previous data suggested an intracellular transit time for newly synthesized acetylcholine receptors of about 3 hr (Devreotes and Fambrough, 1975, 1976b). The density-shift technique was used to verify the precursor-product relationship existing between pool and newly appearing surface receptors. When myotubes are incubated in ^{2}H, ^{13}C, ^{15}N-amino acids, about 50% of the pool receptors are replaced by ^{2}H, ^{13}C, ^{15}N-receptors within 3 hr of addition of the isotopic label. The total number of pool receptors remains constant. In contrast, few ^{2}H, ^{13}C, ^{15}N-receptors appear on the surface during the first 2 hr after introduction of the isotopic label. Slightly over 3 hr after introduction of the isotopic label, most of the new surface receptors incorporated are ^{2}H, ^{13}C, ^{15}N-receptors. These kinetics strongly support the conclusion that receptors undergo an intracellular transit time of 3 hr before being incorporated into the myotube surface.

It is probable that the pool we have characterized includes most of the receptors undergoing intracellular transport. If the pool did not include a large fraction of the receptors in the early stages of transport, a long lag would be expected before newly synthesized receptors could be observed in the pool. As shown in Figure 5, little or no lag is observed. On the other hand, if a large fraction of the receptors in the late stages of transport were absent, we would expect the appearance of ^{2}H, ^{13}C, ^{15}N-receptors in the pool to plateau long before any ^{2}H, ^{13}C, ^{15}N-receptors appeared on the surface. As shown in Figure 7, the rapid appearance of ^{2}H, ^{13}C, ^{15}N-receptors in the pool plateaus at about the same time that ^{2}H, ^{13}C, ^{15}N-receptors begin to appear in large numbers on the surface.

In addition, the pool size quantitatively matches expectation—that is, new receptors are incorporated into the surface at a rate of about 4% per hour (when normalized to the total number of receptors exposed to extracellular α-bungarotoxin). The transport time for receptors through the pool is about 3 hr. This suggests that the pool size should be about 12%. The total pool size in this study is about 20%. As seen in Figure 5, however, only about 50% of these receptors are rapidly replaced by ^{2}H, ^{13}C, ^{15}N-receptors and therefore expected to be precursor receptors. The slow kinetics of replacement of the remaining 50% of the pool by ^{2}H, ^{13}C, ^{15}N-receptors suggest that these receptors might be metabolically equivalent to surface receptors. It is not understood at present why this small fraction of receptors cannot interact with extracellular α-bungarotoxin. The quantitative relationship between the rate of appearance of ^{2}H, ^{13}C, ^{15}N-receptors in the pool and on the surface is illustrated by Figure 7.

One can imagine two extreme models for the mechanism of intracellular receptor transport. Receptors might be transported by a strictly linear assembly-line process. Such a mechanism would lead to linear kinetics of labeling of the pool receptors and an absolute lag before any labeled receptors were incorporated into the surface membrane. On the other hand, the mechanism of transport

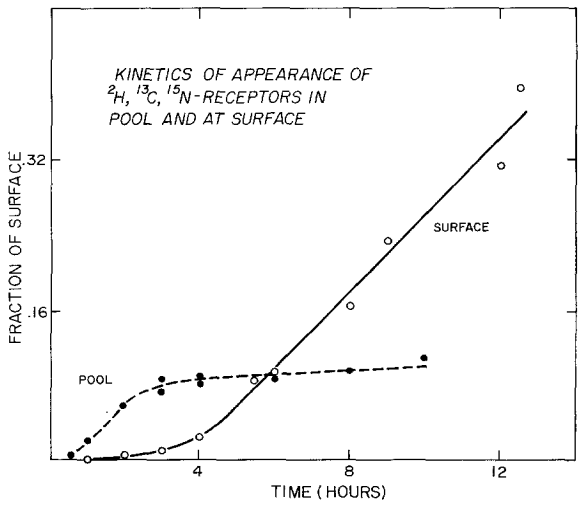

Figure 7. Precursor-Product Relationship of Pool and Surface Receptors

Data from Figures 3 and 5 were plotted together. Pool data have been normalized so that the pool to surface ratio is 0.20, the average value obtained in this study. (●- - -●) kinetics of labeling of pool receptors with ^{2}H, ^{13}C, ^{15}N-amino-acids; (○——○) kinetics of labeling of surface receptors with ^{2}H, ^{13}C, ^{15}N- amino acids.

might be random—all receptors in the pool having equal probability of entering the surface membrane. This mechanism would lead to first-order kinetics for both the "switchover" of pool and the incorporation of newly synthesized receptors into the surface. The observed data are intermediate between the two extreme cases. This might suggest that there exists both linear assembly-line elements and also random elements in the pathway of receptor transport to the surface membrane. The intracellular transport of the digestive enzymes would be expected to exhibit such intermediate kinetics. Transport from the endoplasmic reticulum to the condensing vacuoles is probably an assembly-line process. Once the digestive enzymes are packaged in zymogen granules, however, both newly synthesized and stored enzymes are equally likely to be secreted (Tartakoff et al., 1974; Scheele and Palade, 1975). We cannot completely rule out an assembly-line model for acetylcholine receptors, since the development of myogenic cells in culture is not completely synchronous. Thus the cultures contain myotubes which are heterogeneous with respect to size and to developmental age, and young myotubes, for example, might have shorter (or longer) intracellular receptor transit times than old myotubes. Such heterogeneity would lead to apparent nonlinear kinetics.

Several aspects of intracellular transport of the secretory proteins (reviewed by Palade, 1959, 1975) are similar to some of the characteristics of acetylcholine receptor biosynthesis. For instance, these proteins are in transit for a few hours after synthesis and before their appearance at the cell surface. The movement of protein toward the cell surface in each case is independent of protein synthesis but does depend upon temperature and oxidative phosphorylation. In addition, it appears that in each case, intermixing of newly synthesized and stored proteins might occur at some point in the transit. Finally, since all the secretory proteins and the acetylcholine receptor are glycoproteins (Meunier et al., 1972a; Brockes and Hall, 1975; Raftery et al., 1976), all might be glycosylated in the endoplasmic reticulum and Golgi apparatus. Recent observations indicate that several membrane glycoproteins, rhodopsin in visual receptor cells, cell surface immunoglobulin in B lymphocytes, and vesicular stomatitus viral glycoprotein in infected mammalian cells also undergo lengthy intracellular transit before exposure on the cell surface (Young, 1973; Vitetta and Uhr, 1975; Atkinson, Moyer, and Summers, 1976). These proteins share some of the other characteristics of transport mentioned above. It is reasonable to suppose that each of these "intergral" membrane proteins is synthesized on membrane-bound polysomes and processed by certain elements of the secretory apparatus during transport toward the cell surface (see also Morrison and Lodish, 1975; Lodish and Small, 1975).

Experimental Procedures

Materials

Eagle's minimum essential medium (EMEM) and horse serum were obtained from GIBCO (Grand Island, New York). ^{2}H, ^{13}C, ^{15}N-amino acid mixtures (98%, ^{2}H; 80%, ^{13}C; 80%, ^{15}N), prepared from algal hydrolysates, as well as deuterium oxide were obtained from Merck, Sharp, and Dohme (St. Louis, Missouri). Sodium cholate was purchased from Calibiochem (Gaithersburg, Maryland). All other chemicals were obtained from commercial sources in the United States.

α-Bungarotoxin was purified and iodinated, and the iodinated form was purified and characterized as described in previous publications [Devreotes and Fambrough (1975) and references therein].

Culturing and Media

Chick myogenic cells were prepared by mechanical disrupton, as previously described (Devreotes and Fambrough, 1975). Cells were cultured in Eagle's minimum essential medium containing 10% horse serum, 2% embryo extract, and 18 mM Na bicarbonate (EMEM 210). Medium containing ^{2}H, ^{13}C, ^{15}N-amino acids was prepared as follows. Each 100 ml of medium contained 9 ml of 10× Hanks' balanced salt solution, 1 ml of minimal essential vitamins, 10 ml of horse serum, and 2 ml of embryo extract (both exhaustively dialyzed against Hanks' balanced salt solution), 100 mg of dextrose, 1.5 mg cysteine, 0.4 mg tryptophan, 0.7 mg glutamine, and 50 mg of ^{2}H, ^{13}C, ^{15}N-amino acid mixture. The commercial amino acid mixture was dissolved in water and filtered through an Amicon UM-2 filter before use. When cells differentiated in medium containing the ^{2}H, ^{13}C, ^{15}N-amino acids, myotubes were thinner than controls, and the extent of fusion did not appear to be as great. ^{2}H, ^{13}C, ^{15}N-amino acids had no effect on morphology of myotubes which had fused in normal medium, and the adverse effects of the ^{2}H, ^{13}C, ^{15}N-amino acids on differentiating cells could be reversed by switching to normal medium.

Interaction of α-Bungarotoxin with Receptors and Preparation of Extracts

Some α-bungarotoxin-receptor complexes were prepared by interaction of α-bungarotoxin with intact living myotubes. Cultures were usually incubated with 0.1 μg/ml ^{125}I- or ^{131}I-labeled α-bungarotoxin in EMEM 210 for 20-30 min at 37°C. To stop the reaction, the medium was removed and cultures were rinsed 3 times for 8-10 min each. When large (100 mm diameter) cultures were used, about 20 ml of wash medium were poured onto the culture and later removed by inverting the culture dishes over a sink. When small cultures (35 mm) were used, cultures were immersed in a large bath of wash medium and removed during changes of the bath medium. The wash medium was Hanks' balanced salt solution at pH 7.2 containing 0.5% BSA. Unless otherwise stated, rinsing was carried out at 4°C. The mono-iodo-α-bungarotoxin-receptor complexes were then extracted from the cells with a solution containing 2.5% sodium cholate (w/v) and 10 mM Tris-HCl buffer (pH 7.8).

The saturation procedure described above blocks cholinergic responses in the muscle cultures (Hartzell and Fambrough, 1973) but leaves another class of receptors, termed "hidden" receptors, unbound (Devreotes and Fambrough, 1975, 1976b). The hidden receptors interact more slowly with α-bungarotoxin, probably due to their location on plasma membrane vesicles or complicated infoldings of the plasma membrane. In cases where the pool receptors were to be studied as complexes with iodine-labeled α-

bungarotoxin, the surface and hidden receptor sites were first blocked by incubating the muscle cultures overnight at 37°C in medium containing 0.1–0.5 μg/ml unlabeled α-bungarotoxin. The receptors which were not complexed with α-bungarotoxin during this incubation presumably represent receptors which are completely unavailable for α-bungarotoxin binding, due to their intracellular location. A major component of this receptor pool is the precursor-receptor population. Receptors in the pool population interact with α-bungarotoxin after solubilization in nonionic detergent solution.

Detergent extracts were usually prepared by one of the two following procedures. Small cultures (35 mm dishes) were usually extracted directly into detergent solution. 1 ml of 1% Triton X-100 or 1% sodium cholate, 10 mM Tris (pH 7.8) was placed directly in the culture dishes, and receptors were solubilized with vigorous pipetting. After removal of the detergent solution containing most of the receptors or α-bungarotoxin-receptor complexes, cultures were reextracted with 0.5 ml of detergent solution. The two extracts were combined. Large cultures (100 mm dishes) were used when we wished to extract many more receptors into a small volume. Each large culture was covered with 2 ml of 10 mM Tris (Ph 7.8). After several minutes, the myotubes were scraped from the dish with a plastic file tab (Oxford Pendaflex, Garden City, New York). The myotubes from about 3–8 dishes were combined in a Dounce homogenizer. Cells were disrupted with 3–5 strokes, and membranes were collected by centrifugation at 27,000 g for 1 hr. The membrane pellet was extracted with 0.5 ml of 2.5% sodium cholate, 10 mM Tris (pH 7.8) by vigorous vortexing for 60 sec. Insoluble material was removed by centrifugation at 27,000 g for 10 min. The final supernatant contained about 80% of the receptors or α-bungarotoxin-receptor complexes originally on the myotubes.

Soluble receptors in detergent solution were labeled with α-bungarotoxin under the following condition, and the complexes formed were separated from free α-bungarotoxin by BioGel P60 chromatography. Nonspecific binding of α-bungarotoxin to detergent extracts of chick myogenic cultures has been an obstacle in reliable preparation of soluble α-bungarotoxin receptor complexes from precursor pools (Devreotes and Fambrough, 1975). Nonspecific binding is characterized as α-bungarotoxin binding which emerges in the void volume of BioGel P60 columns and sediments in the 2S-6S region of sucrose gradients. Binding of α-bungarotoxin to the nonspecific sites is not d-tubocurarine-competed. It presents a range of reaction rates, some of which are higher than for α-bungarotoxin-receptor complexes. Methods have been empirically derived which eliminate or greatly reduce the amount of nonspecific binding. Extracts in 2.5% sodium cholate, 10 mM Tris (pH 7.8) were preincubated at 37°C for 2–3 hr. α-Bungarotoxin was then added to the extract at a final concentration of 0.01 μg/ml (1.25 nM), and incubation was continued at 37°C for an additional 20–30 min. The reaction was terminated by loading the reaction mixture onto BioGel P60 columns equilibrated with 1% sodium cholate, 10 mM Tris (pH 7.8). α-Bungarotoxin-receptor complexes were freed of unbound α-bungarotoxin by chromatography on 13 ml BioGel P60 columns poured in 10 ml disposable plastic pipettes. Under these conditions, about 70–100% of the radioactivity emerging in the void volume in typical pool assays was α-bungarotoxin-receptor complexes (defines as 10S material by velocity sedimentation in sucrose gradients). The preincubation at 37°C resulted in a great reduction in the amount of nonspecific binding with no detectable loss in receptor activity—that is, reactions run at 22°C with no preincubation yielded the same number of α-bungarotoxin-receptor complexes but a great deal more nonspecific binding.

Sucrose Gradients for Estimation of the Magnitude of the Density Shift

An estimate of magnitude of the density shift of the receptor was made as follows. The rates of sedimentation of mono-iodo-α-bungarotoxin-receptor complexes involving normal and density-shifted receptors were determined in gradients of various density. The data were then extrapolated to find the density at which the sedimentation rate of each complex would be zero. For sufficiently shallow gradients, the rate of migration is approximately constant (see Martin and Ames, 1961). Under these conditions, the distance traveled by a macromolecule is given by: $X = X_0 \lambda t$, where X is the distance between the position of the macromolecule and the center of rotation at time t, and X_0 is the distance from the meniscus to the center of rotation. λ is given by the following expression: $\lambda = (K/\rho)(\rho - \rho_0)$, where K is a constant dependent on the rotor speed, temperature, and molecular weight; ρ_0 is the average density of the shallow gradients; and ρ is the density of the macromolecule. Thus linear plots of λ vs ρ_0 intersect the ρ_0 axis at ρ. Shallow gradients of different average densities were prepared by substituting D_2O for H_2O. The shallowest gradient which did not allow thermal convection and consequent band broadening was found to be 5–20% sucrose. Continuous 5–20% sucrose gradients were prepared using H_2O, 50% D_2O, and 100% D_2O as solvents. The gradients also contained 1% Triton X-100, 10 mM Tris (pH 7.8), 1 mM EDTA, and 1 mM phenyl methane sulfonyl flouride (PMSF). The initial and final densities of the gradients were 1.01 g/ml and 1.03 g/ml for H_2O, 1.05 g/ml and 1.07 g/ml for 50% D_2O, and 1.10 and 1.13 g/ml for D_2O. All centrifugation was carried out in a Beckman SW50.1 rotor at 48,000 rpm at 4°C. Acceleration (3 min) and deceleration (3 min) times were omitted from the calculations.

Separation of ^2H, ^{13}C, ^{15}N-Receptors and ^1H, ^{12}C, ^{14}N-Receptors and Analysis of Gradient Profiles to Determine the Proportion of ^2H, ^{13}C, ^{15}N- and ^1H, ^{12}C, ^{14}N-Receptors

Maximum separation of α-bungarotoxin-^2H, ^{13}C, ^{15}N-receptor complexes and α-bungarotoxin-^1H, ^{12}C, ^{14}N-receptor complexes was achieved by velocity sedimentation in sucrose gradients. Sucrose-deuterium oxide gradients were formed in cellulose nitrate centrifuge tubes by layering of solutions of the following sucrose concentrations: 4.0 ml of 40% (w/v), 4.0 ml of 32.5% (w/v), 3.5 ml of 25% (w/v). The function of the gradient is to stabilize the solution against convection currents. The small discontinuities in density probably dissipate during the long centrifugation. Deuterium oxide and high concentrations of sucrose were used so that the average density of the gradients would be high. This increases the resolving power (see Figure 2, right panel). Centrifugation was performed in a Beckman SW41 rotor at 40,000 rpm at 4°C.

As demonstrated in this report, receptors extracted from myotubes grown in ^2H, ^{13}C, ^{15}N-amino acid medium are of about the same density as ^2H, ^{13}C, ^{15}N-receptors synthesized by differentiating myotubes (see Figures 1, 4, and 6). In addition, it appears that very few receptors of intermediate density are produced—that is, in the course of these experiments, we have observed only two classes of receptors. Accordingly, we have analyzed the gradient data where two peaks were present with a computer program which adds different proportions of ^2H, ^{13}C, ^{15}N- and ^1H, ^{12}C, ^{14}N-receptors.

^{131}I-mono-iodo-α-bungarotoxin-^1H, ^{12}C, ^{14}N-receptor complexes ("markers") were mixed with the samples to ascertain the position and shape of the ^{125}I-mono-iodo-α-bungarotoxin-^1H, ^{12}C, ^{14}N-receptor complex component of the sample in the sucrose gradient. Using the "marker" peak as a "template," the proportions of the ^{125}I-mono-iodo-α-bungarotoxin-^2H, ^{13}C, ^{15}N-receptor complexes and ^{125}I-α-bungarotoxin-^1H, ^{12}C, ^{14}N-receptor complexes on each gradient were estimated. Before sucrose gradients of precursor pool receptors were analyzed, the contribution of the nonspecific binding, shown in Figure 6, to the receptor peaks was estimated by extrapolating the leading edge of the nonspecific binding peak to the base line. This correction was largest for cases where there was little ^1H, ^{12}C, ^{14}N-receptor (that is, longest time points). In any event, these corrections were small. The correction to the 10 hr point resulted in a 7% change in the estimated proportion of ^1H, ^{12}C, ^{14}N-receptors in the sample.

Theoretical superposition of "heavy" and "light" peaks (each having the shape of the marker) of different proportions was carried out with the aid of a DEC PDP-8 computer. The computer simultaneously plotted the real data and the theoretically constructed double peak. The computer operator could repeat this process, each time adjusting the theoretical proportion of "heavy" and "light" peaks. The best fit was obtained by visual inspection. The error bars in Figures 3 and 5 indicate the limits on each side of the best fit where it was clear that the theoretically constructed curve did not superimpose on the real curve. When samples were prepared by mixing known proportions of heavy and light receptors, and this type of analysis was carried out, there was extremely good correlation between known and calculated values.

S Value of Receptors

In the course of the experiments reported above, we have repeatedly determined the sedimentation coefficient of α-bungarotoxin-receptor complexes. The S value of α-bungarotoxin-receptor complexes has been extensively reported to be 9S-10S, and we find that the S value of α-bungarotoxin-receptor complexes extracted from embryonic chick myotubes is, similiarly, about 10S (Karlin, 1974; Devreotes and Fambrough, 1975). We noticed, however, that the manner in which the α-bungarotoxin-receptor complexes are formed and handled can have an effect on the S value. For instance, when α-bungarotoxin-receptor complexes are formed on intact myotubes and subsequently extracted in nonionic detergents, the S value of the complexes is about 10S. On the other hand, if receptors are first extracted into nonionic detergents and reacted with α-bungarotoxin in solution, the S value of the complexes is about 9.5S. Preliminary evidence also suggests that heat treatment of 10S complexes converts them to 9.5S complexes. We do not know the significance of this small shift or whether the shifts produced by the various treatments are equivalent. However, comparisons of α-bungarotoxin-receptor complexes formed or handled by different protocols can lead to artifacts. No difference is S value has been observed among the α-bungarotoxin-receptor complexes formed from surface, "hidden," pool, and newly incorporated surface receptors as long as the complexes were formed and handled identically. Accordingly, when comparisons were made between ^{125}I-mono-iodo-α-bungarotoxin-receptor complexes and ^{131}I-mono-iodo-α-bungarotoxin-receptor complexes, each type of complex was formed in an identical manner.

Radioisotope Counting

In most cases, both ^{131}I- and ^{125}I-mono-iodo-α-bungarotoxin-receptor complexes were run on the same gradients. Fractions (about 0.2 ml) were collected into scintillation vials, and 3.5 ml of scintillation cocktail (Devreotes and Fambrough, 1975) were added. The activity of each isotope was determined simultaneously by using two appropriately adjusted channels of a Packard Tricarb scintillation spectrometer.

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